

Tran Trong Nhan

Software Engineer | Backend & AI

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Education

FPT Academy (Aptech), Advanced Diploma in Software Engineering – Ho Chi Minh City, Vietnam

2023 – 2026

- Graduated with Distinction (84.25/100)

Projects

recon-qa – Open-Source QA & Automation Engine | [PyPI](#) | [GitHub](#)

Aug 2026 – Present

Python · FastAPI · Playwright · Pydantic · SQLAlchemy · SQLite · PyPI

- Published a QA automation package on PyPI (`pip install recon-qa`) that discovers OpenAPI specs and crawls web apps with Playwright to automatically generate API and UI test suites.
- Built an asynchronous execution engine with bounded worker pools supporting HTTP and Playwright runners, automated failure screenshots, and self-contained HTML reports.
- Integrated LLM-based exploratory test generation and dual-layer failure classification with multi-provider root-cause analysis (Gemini, Mistral, OpenAI, Ollama) to generate actionable code-fix recommendations.

Groundwork – AI Document Workspace | [Live Demo](#) | [GitHub](#)

June 2026 – Aug 2026

FastAPI · Python · React · PostgreSQL · pgvector · Celery · Redis · Docker

- Architected a full-stack document workspace using FastAPI, React, and PostgreSQL/pgvector for multi-format document ingestion, semantic chunking, and vector retrieval.
- Orchestrated asynchronous OCR and AI generation workflows via Celery/Redis background jobs with per-job cancellation, deployed to Google Cloud Run.
- Built citation verification that cross-checks claims against source pages and prevents ungrounded exports, backed by 102 Pytest tests and Playwright E2E suites.

Tenvora – Enterprise Payment Operations & Ledger | [Live Demo](#) | [GitHub](#)

Mar 2026 – May 2026

ASP.NET Core · C# · React · TypeScript · EF Core · PostgreSQL · Docker

- Architected a multi-tenant payment operations and double-entry ledger platform with pessimistic row-locking (`SELECT FOR UPDATE`), gross-to-net settlement batches, and automated bank reconciliation.
- Enforced multi-tenant workspace isolation using JWT tenant scoping and EF Core interceptors with tamper-evident JSON audit logging.
- Built variance threshold matching, rate limiting, and asynchronous central bank exchange rate synchronization, verified by 16 xUnit tests and Vitest suites.

LogiFlow – Freight Management Platform | [Live Demo](#) | [GitHub](#)

Oct 2025 – Jan 2026

Academic Team Project · Team Lead · Spring Boot · Java · React · Flutter · PostgreSQL · Docker

- Led a 4-member team and built the core backend, including PostgreSQL schema, authentication, authorization, and multi-role logistics workflows with Spring Boot.
- Developed real-time driver tracking with WebSockets and React/Leaflet, plus proof-of-delivery signature capture and live route monitoring.
- Integrated OCR-based driver verification, constraint-based driver assignment, PayPal checkout, and PDF invoicing; containerized with Docker Compose and GitHub Actions CI.

Technical Skills

Languages: Python, C#, Java, TypeScript, JavaScript, SQL, Dart

Backend & Frameworks: FastAPI, ASP.NET Core, Spring Boot, REST APIs, WebSockets, Celery

Frontend & Mobile: React, Flutter, HTML5/CSS3, Tailwind CSS

AI/LLM: OpenAI, Gemini, Mistral, Ollama, RAG

Databases & Storage: PostgreSQL, pgvector, Redis, S3/MinIO

Infrastructure & CI/CD: Docker, Google Cloud Run, GitHub Actions

Testing & Tools: Pytest, xUnit, JUnit, Playwright, Vitest, Git, Postman